# Exercise 3: Sensors & Actuators

## Introduction

In this exercise, you will build a pneumatic system consisting of two air pumps, an air valve, and an inflatable pillow — and bring it to life with a sensor-driven interaction of your own design.

The pneumatic circuit is straightforward: one pump inflates the pillow, the other deflates it, and an air valve switches between the two. Control is handled by MOSFET driver modules, which let your Arduino switch the pumps and valve on and off safely. Your creative task is to choose how a person interacts with this system — what sensor or input method triggers inflation or deflation, and what kind of experience that creates.

**Record your work as you go —** take photos of each circuit configuration and note down all observations. You will need this material when writing up your portfolio report. Record a video of the final working system.

> **Important:** This instruction does not provide a step-by-step recipe. It explains the pneumatic and electronic components, and leaves the design of interaction and sensor integration to you. If in doubt about any component, check the linked product pages or datasheets first — then ask the tutor.

## Few General Rules

- **Power the pumps and valve from the lab power supply only.** The microcontroller and all sensors must be powered from USB.
- **Set current limits carefully.** Watch the current draw on the power supply closely. If the draw reaches hundreds of milliamps with no motors spinning, cut the power immediately.
- **Always switch off and disconnect power before modifying or rewiring a circuit.** Never change connections while the circuit is live.
- **You cannot drive the pumps or valve directly from Arduino pins.** Use the provided MOSFET modules — they are rated for the current required.
- **Ensure all parts of your circuit share a common ground.** The Arduino GND and the power supply GND must be connected.
- **Double-check polarity before powering on.** Connecting a component backwards can damage it permanently.
- If anything smells, sparks, or gets unexpectedly hot — immediately disconnect power and inform the tutor.
- **If you are unsure about anything at any point, ask the tutor before powering your circuit.**

# The Exercise

The exercise has two parallel threads that you will eventually combine. Work through them in whichever order suits you.

## Part A — Pneumatic & Electrical Circuit

Assemble and verify the actuator side of the system:

1. **Assemble the electrical circuit** for controlling 2 pumps and 1 valve using 3 MOSFET modules. Wire the power supply to the load side of the modules and connect the control side to Arduino digital pins.
2. **Verify operation** by creating and uploading a test sketch that turns each actuator on and off in a looped sequence. Use the MOSFET module's built-in status LED to confirm correct switching — if the LED lights but the pump doesn't run, check your load-side wiring and power supply.
3. **Assemble the pneumatic circuit** by connecting the pumps, valve, and inflatable pillow with silicone tubing. Pay attention to the valve port assignments.
4. **Test the complete system** — inflate and deflate the pillow from your test code before moving on.

## Part B — Sensor Interaction

Choose a sensor or input method and develop the interaction:

- Think about which sensors and interactions you would like to explore — browse what is available in the components storage.
- Pick your chosen sensor and look up its datasheet and any relevant tutorials online.
- Install any required libraries in the Arduino IDE.
    - Most libraries come with **example sketches — use them as a starting point.**
- Wire the sensor to the Arduino and experiment with it. Think about how you want the interaction to feel.

> **Example shown during class:** An ultrasonic distance sensor detects a hand moving downward (towards the floor), mimicking the action of a bicycle pump. Pushing down inflates the pillow; releasing deflation via a button press mimics releasing air from a tire valve. Use this as inspiration — or come up with something entirely different.

## Combining Both Parts

Once you have a working actuator system and a working sensor input, combine them: use your sensor readings to trigger inflation or deflation of the pillow. This is the core of the exercise.

# Components Overview

The exercise uses three actuator components — two pumps and one valve — all controlled via MOSFET modules from the Arduino. The most important specifications are listed below. Product pages and the datasheet are linked for reference.

## Air Pump — ZR370-02PM (×2)

Product page: Adafruit #4699 — Air Pump and Vacuum DC Motor 4.5V / 2.5 LPM

Two of these pumps are used in the exercise — one for inflation, one for deflation. Key things to know:

| Air Pump ZR370-02PM — Key Specifications | |
|---|---|
| **Operating voltage** | ~4.5 V (5 V is fine) |
| **Current draw** | ~500 mA at operating voltage |
| **Connector** | Bare wire leads — positive terminal marked with a red dot. A protection diode is soldered to the motor leads (back-EMF protection). **Polarity must not be reversed.** Use the provided alligator clips to connect. |
| **Rated use** | Not rated for continuous use. Specification sheet indicates the design is tested for 30,000 cycles of 10 seconds on, 5 seconds off. Recommended duty cycle: ~50%. |

**⚠ The positive terminal of each pump is marked with a red dot. A diode has been soldered to the motor leads to provide back-EMF protection — because of this, the polarity must not be reversed. Use the provided alligator clips to connect to the pump leads.**

The pump draws air in through the side of the plastic casing and pushes it out through the tubing port. When powered off, the pump holds air (no passive leaking through the pump itself).

## Air Valve — FA0520E (×1)

Product page: Adafruit #4663 — 6V Air Valve with 2-pin JST XH Connector

Think of the valve as a pneumatic relay: it has three ports and switches the air path depending on whether it is powered.

| Air Valve FA0520E — Key Specifications | |
|---|---|
| **Operating voltage** | 6 V (works fine at 5 V) |
| **Current draw** | ~400 mA |
| **Connector** | 2-pin JST XH |
| **Number of ports** | 3 ports (one common + two switching) |

| | |
|---|---|
| **Unpowered state** | Common port connected to the metal-end port (normally open path) |
| **Powered state** | Common port switches to the plastic-end port (the other path closes) |
| **Application** | Switching between inflation and deflation pump paths |

> **Port wiring:** The middle port (flanged) is the common connection. When the valve is unpowered, the common port and the metal-end port are connected. When powered, the common port and the plastic-end port are connected, and the metal-end port is closed. Plan your pneumatic circuit around this switching behaviour.

## MOSFET Driver Module — IRF520 (×3)

Product page: Reichelt — MOS Driver Module 3.3–5V / 0–24V IRF520 (ST1168)
Datasheet: ST1168 Datasheet (PDF)

One module is used per actuator (two pumps + one valve = three modules). Each module lets a single Arduino digital pin switch a high-current DC load. It also includes a built-in status LED that turns on when the MOSFET is conducting — very useful for debugging.

| IRF520 MOSFET Module — Key Specifications | |
|---|---|
| **Control signal voltage** | 3.3 V or 5 V (Arduino compatible) |
| **Load voltage (V+ / V−)** | 0–24 V DC |
| **Max load current** | Up to 5 A (add heatsink above ~1 A) |
| **Status LED** | On when SIG is HIGH (load is switched on) |
| **Logic** | SIG HIGH → load ON; SIG LOW → load OFF |
| **Connector — control side** | 3-pin header: GND, VCC, SIG |
| **Connector — load side** | Screw terminals: V+, V− |
| **Power supply** | Separate from microcontroller (connect to lab power supply) |

> **Wiring the module:** Connect the module's GND pin to Arduino GND. Connect SIG to any Arduino digital output pin. Connect the load (pump or valve) to the V+ / V− screw terminals. Connect the separate power supply (for the pumps/valve) to Vin and GND on the module's load side. The module's built-in LED will light up whenever SIG goes HIGH.

# Portfolio Report

Document your process as you go. The portfolio should include:

- **Description of steps taken** during the exercise — what you built, in what order, and how.
- **What worked and what failed** — honest reflection on problems encountered and how you solved (or didn't solve) them.
- **Photos and/or videos** of the circuits, pneumatic assembly, and the final working interaction.
- **Sensor / interaction design explanation:** which sensor or input method did you choose and why? Possible motivations:
    - Exploring how a sensor you found interesting works.
    - Playful interaction — mimicking a real-world action or gesture.
- **Resources used:** which datasheets, tutorials, or libraries did you rely on for your sensor? Was finding this information straightforward or challenging?
- **Reflection on the outcome:** did you achieve your goal? Does the final interaction match your initial idea? Are there other approaches you would like to explore? Was it fun, challenging, easy, or engaging?